

Fifth Degree Function Report

Polynomial

$$f(q) = aq^5 + bq^4 + cq^3 + dq^2 + eq + f$$

Naive

$$g_{\text{naive}}(X) = X^5a + X^4b + X^3c + X^2d + Xe + f$$

$$\begin{aligned}\mathbb{E}[g_{\text{naive}}(X)] &= aq^5 + bq^4 + 24\frac{\Delta^4}{\epsilon} (5aq + b) \\ &\quad + 2\frac{\Delta^2}{\epsilon} (10aq^3 + 6bq^2 + 3cq + d) + cq^3 + dq^2 + eq + f\end{aligned}$$

$$\begin{aligned}\text{Var}(g_{\text{naive}}(X)) &= 2\frac{\Delta^2}{\epsilon} \left(1814400a^2\frac{\Delta^8}{\epsilon} + 25a^2q^8 + 40abq^7 + 30acq^6 + 20adq^5 \right. \\ &\quad + 10aeq^4 + 16b^2q^6 + 24bcq^5 + 16bdq^4 + 8beq^3 \\ &\quad + 288\frac{\Delta^6}{\epsilon} (3125a^2q^2 + 1250abq + 140ac + 69b^2) \\ &\quad + 24\frac{\Delta^4}{\epsilon} (3050a^2q^4 + 2440abq^3 + 810acq^2 + 200adq + 30ae \\ &\quad + 408b^2q^2 + 204bcq + 28bd + 15c^2) + 2\frac{\Delta^2}{\epsilon} (1100a^2q^6 \\ &\quad + 1320abq^5 + 720acq^4 + 340adq^3 + 120aeq^2 + 372b^2q^4 \\ &\quad + 372bcq^3 + 156bdq^2 + 48beq + 81c^2q^2 + 54cdq + 12ce + 5d^2) \\ &\quad \left. + 9c^2q^4 + 12cdq^3 + 6ceq^2 + 4d^2q^2 + 4deq + e^2 \right)\end{aligned}$$

$$\begin{aligned}\text{MSE}(g_{\text{naive}}) &= 2\frac{\Delta^2}{\epsilon} \left(1814400a^2\frac{\Delta^8}{\epsilon} + 25a^2q^8 + 40abq^7 + 30acq^6 + 20adq^5 \right. \\ &\quad + 10aeq^4 + 16b^2q^6 + 24bcq^5 + 16bdq^4 + 8beq^3 \\ &\quad + 20160\frac{\Delta^6}{\epsilon} (45a^2q^2 + 18abq + 2ac + b^2) + 360\frac{\Delta^4}{\epsilon} (210a^2q^4 \\ &\quad + 168abq^3 + 56acq^2 + 14adq + 2ae + 28b^2q^2 + 14bcq + 2bd + c^2) \\ &\quad + 12\frac{\Delta^2}{\epsilon} (200a^2q^6 + 240abq^5 + 130acq^4 + 60adq^3 + 20aeq^2 \\ &\quad + 68b^2q^4 + 68bcq^3 + 28bdq^2 + 8beq + 15c^2q^2 + 10cdq + 2ce + d^2) \\ &\quad \left. + 9c^2q^4 + 12cdq^3 + 6ceq^2 + 4d^2q^2 + 4deq + e^2 \right)\end{aligned}$$

$$\text{Bias}(g_{\text{naive}}) = 2\frac{\Delta^2}{\epsilon} \left(10aq^3 + 6bq^2 + 12\frac{\Delta^2}{\epsilon} (5aq + b) + 3cq + d \right)$$

Unbiased

$$g_{\text{unb}}(X) = X^5a + X^4b + X^3c + X^2d + Xe - 2\frac{\Delta^2}{\epsilon} (10X^3a + 6X^2b + 3Xc + d) + f$$

$$\mathbb{E}[g_{\text{unb}}(X)] = aq^5 + bq^4 + cq^3 + dq^2 + eq + f$$

$$\begin{aligned} \text{Var}(g_{\text{unb}}(X)) = 2\frac{\Delta^2}{\epsilon} & \left(1152000a^2\frac{\Delta^8}{\epsilon} + 25a^2q^8 + 40abq^7 + 30acq^6 + 20adq^5 \right. \\ & + 10aeq^4 + 16b^2q^6 + 24bcq^5 + 16bdq^4 + 8beq^3 \\ & + 288\frac{\Delta^6}{\epsilon} (2000a^2q^2 + 800abq + 85ac + 46b^2) \\ & + 12\frac{\Delta^4}{\epsilon} (4000a^2q^4 + 3200abq^3 + 1020acq^2 + 220adq + 20ae \\ & + 552b^2q^2 + 276bcq + 36bd + 21c^2) + 2\frac{\Delta^2}{\epsilon} (800a^2q^6 + 960abq^5 \\ & + 510acq^4 + 220adq^3 + 60aeq^2 + 276b^2q^4 + 276bcq^3 + 108bdq^2 \\ & + 24beq + 63c^2q^2 + 42cdq + 6ce + 5d^2) + 9c^2q^4 + 12cdq^3 \\ & \left. + 6ceq^2 + 4d^2q^2 + 4deq + e^2 \right) \end{aligned}$$

$$\begin{aligned} \text{MSE}(g_{\text{unb}}) = 2\frac{\Delta^2}{\epsilon} & \left(1152000a^2\frac{\Delta^8}{\epsilon} + 25a^2q^8 + 40abq^7 + 30acq^6 + 20adq^5 \right. \\ & + 10aeq^4 + 16b^2q^6 + 24bcq^5 + 16bdq^4 + 8beq^3 \\ & + 288\frac{\Delta^6}{\epsilon} (2000a^2q^2 + 800abq + 85ac + 46b^2) \\ & + 12\frac{\Delta^4}{\epsilon} (4000a^2q^4 + 3200abq^3 + 1020acq^2 + 220adq + 20ae \\ & + 552b^2q^2 + 276bcq + 36bd + 21c^2) \\ & + 2\frac{\Delta^2}{\epsilon} (800a^2q^6 + 960abq^5 + 510acq^4 + 220adq^3 + 60aeq^2 + 276b^2q^4 \\ & + 276bcq^3 + 108bdq^2 + 24beq + 63c^2q^2 + 42cdq + 6ce + 5d^2) \\ & \left. + 9c^2q^4 + 12cdq^3 + 6ceq^2 + 4d^2q^2 + 4deq + e^2 \right) \end{aligned}$$

Comparison

$$\text{mean gap} = -2\frac{\Delta^2}{\epsilon} \left(10aq^3 + 6bq^2 + 12\frac{\Delta^2}{\epsilon} (5aq + b) + 3cq + d \right)$$

$$\begin{aligned}
\text{variance gap} = & -24 \frac{\Delta^4}{\epsilon} \left(55200a^2 \frac{\Delta^6}{\epsilon} + 50a^2q^6 + 60abq^5 + 35acq^4 + 20adq^3 \right. \\
& + 10aeq^2 + 16b^2q^4 + 16bcq^3 + 8bdq^2 + 4beq \\
& + 24 \frac{\Delta^4}{\epsilon} (1125a^2q^2 + 450abq + 55ac + 23b^2) \\
& + \frac{\Delta^2}{\epsilon} (2100a^2q^4 + 1680abq^3 + 600acq^2 + 180adq + 40ae \\
& \left. + 264b^2q^2 + 132bcq + 20bd + 9c^2) + 3c^2q^2 + 2cdq + ce \right)
\end{aligned}$$

$$\begin{aligned}
\text{MSE gap} = & -4 \frac{\Delta^4}{\epsilon} \left(331200a^2 \frac{\Delta^6}{\epsilon} + 400a^2q^6 + 480abq^5 + 270acq^4 + 140adq^3 \right. \\
& + 60aeq^2 + 132b^2q^4 + 132bcq^3 + 60bdq^2 + 24beq \\
& + 144 \frac{\Delta^4}{\epsilon} (1150a^2q^2 + 460abq + 55ac + 24b^2) \\
& + 6 \frac{\Delta^2}{\epsilon} (2300a^2q^4 + 1840abq^3 + 660acq^2 + 200adq + 40ae + 288b^2q^2 \\
& \left. + 144bcq + 24bd + 9c^2) + 27c^2q^2 + 18cdq + 6ce + d^2 \right)
\end{aligned}$$

$$\begin{aligned}
\text{Bias}(g_{\text{naive}})^2 = & 4 \frac{\Delta^4}{\epsilon} \left(100a^2q^6 + 120abq^5 + 60acq^4 + 20adq^3 + 36b^2q^4 + 36bcq^3 \right. \\
& + 12bdq^2 + 144 \frac{\Delta^4}{\epsilon} (25a^2q^2 + 10abq + b^2) \\
& + 24 \frac{\Delta^2}{\epsilon} (50a^2q^4 + 40abq^3 + 15acq^2 + 5adq + 6b^2q^2 + 3bcq + bd) \\
& \left. + 9c^2q^2 + 6cdq + d^2 \right)
\end{aligned}$$